The Session 4 Operational Map — From Gauge Integers to Cosmological Predictions

The beta coefficients run further than expected. Here is where they go.

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Abstract

Session 3 established the operational ground: the SM does not unify (gap ratio 218/115 vs 1.358, 40% miss), the Cabibbo Doublet (3,2,1/6) fixes it (gap ratio 38/27, distance 0.049), and the two-loop correction improves the unification miss by 66%. Session 4 discovered that the same beta coefficient integers controlling unification also appear to control cosmological observables. Six formulas using only the electromagnetic coupling α, the beta-derived integers 13, 19, 20, and 22, and the geometric constant π predict seven cosmological quantities — the cosmological constant scale, the dark matter to baryon ratio, the Hubble constant, and the four cosmic energy density fractions — at sub-percent precision with zero cosmological input. This paper documents the discovery, sets the working direction, and specifies the research program to determine whether the formulas are physics or coincidence. Three tracks: unification completion (PHYS-26 through PHYS-30), beta cosmology (PHYS-31 through PHYS-35, gated by statistical control), and structural foundations (PHYS-36, PHYS-37). Each paper has a backing script, a verification standard, and an abort condition. The direction is set without hedge. If it is wrong, we backtrack.

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1. Where We Stand

PHYS-24 drew the operational boundary: 329/329 checks, zero failures, everything fixed until falsified. This paper does not re-argue that ground. The gap ratio is 218/115. The Cabibbo Doublet gives 38/27. The two-loop Delta is −0.40. The Koide C_3 path is closed. The PSLQ null is 82/82. DATA-4 has 146 entries at 38/38. These are facts of the series, cited by script name and check count.

Session 4 added three exploration scripts: qed_predicts_gr.py (10/10), qed_gr_scan_2.py (10/10), and beta_unification_test.py (15/15). Combined with the PHYS-24 base: 364/364 checks, zero failures. The exploration scripts found something PHYS-24 did not anticipate: the beta coefficient integers that control gauge coupling unification also produce sub-percent matches to cosmological observables.

PHYS-24 deprioritized "generic cosmological boundary speculation without a derived per-transit law." That deprioritization was correct at the time — no specific formulas existed. The formulas now exist. They pass 15/15 checks. They use only library values. This paper sets the direction for investigating them.

(Backed by phys25_platform.py Section 1: 5/5 checks.)

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2. The Integer Inventory

The beta coefficients of SU(3) × SU(2) × U(1) produce a specific set of integers. These are Level 1 — determined by the gauge group, not by measurement. A civilization with a different value of α but the same gauge group would compute the same integers.

The SM SU(2) beta $b_2 = -19/6$ has numerator magnitude 19. The Cabibbo Doublet adds $\Delta b_2 = 1$, giving $b_2' = -13/6$ with numerator magnitude 13. The modified SU(3) beta $b_3' = -20/3$ has numerator magnitude (times 3) equal to 20. Twice the Yang-Mills integer gives 22 = 2 × 11. The generation count gives 3. Products: 3 × 19 = 57, 3 × 13 = 39.

Two exact algebraic identities connect these integers:

57/39 = 19/13. The ratio of the SM and VL cosmological constant exponents equals the ratio of the SU(2) beta numerators. Verified exact in Fraction arithmetic.

$20/13 = |3b_3'|/|6b_2' \times 6|$. The ratio of the modified SU(3) and SU(2) beta numerators. Verified exact.

These are not numerical observations. They are consequences of the library values through exact Fraction cancellation. They hold in any universe with the same gauge group and particle content.

(Backed by phys25_platform.py Section 2: 8/8 checks, all EXACT.)

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3. The Normalization Resolution

The convention discrepancy from the Session 3 script sin2_theta_w_0.py is resolved. The GUT normalization factor for U(1) is (3/5), entering the Dynkin formula as $\Delta b_1 = (2/5) \times \dim(R_3) \times \dim(R_2) \times Y^2$. For the Cabibbo Doublet (3,2,1/6): $\Delta b_1 = (2/5) \times 3 \times 2 \times (1/6)^2 = (2/5) \times (1/6) = 1/15$.

The MSSM gate verifies the convention: applying the same formulas to the full MSSM particle content reproduces the known MSSM gap ratio 7/5 = 1.400 exactly.

Two independent routes arrive at 1/15. Route A: $(2/5) \times 3 \times 2 \times (1/36) = (2/5) \times (6/36) = (2/5) \times (1/6) = 2/30 = 1/15$. Route B: a vector-like pair counts as 2 complex scalars for U(1), each contributing 1/30, giving $2 \times (1/30) = 1/15$. Both routes verified exact.

(Backed by phys25_platform.py Section 3: 4/4 checks, all EXACT.)

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4. The Cosmological Constant from Beta Exponents

The electromagnetic coupling raised to the power $3 \times |b_2$ numerator$|$ produces the scale of the cosmological constant in Planck units.

Version A (SM beta): α^57 = 10^−121.80. The measured Λ in Planck units is 10^−121.54. The miss is 0.26 decades over 122 orders of magnitude.

Version B (VL beta): (α/(3 π))^39 = 10^−121.33. The miss is 0.21 decades.

The measured value sits between the two predictions: $-121.80 < -121.54 < -121.33$. The interpolation fraction is f = 0.44 — the measured Λ is 44% of the way from the VL prediction to the SM prediction.

The exact identity 57/39 = 19/13 connects the two versions: the ratio of exponents equals the ratio of SU(2) beta numerators before and after the Cabibbo Doublet.

No derivation exists for why α^(3 × $|b_2$ num$|$) should produce the cosmological constant scale. The formula is stated as found, not as derived. The working direction: treat this as the Λ formula and investigate whether the two-loop correction to the effective b_2 closes the 0.2-decade gap.

(Backed by phys25_platform.py Section 4: 4/4 checks.)

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5. The Dark Matter Ratio

The dark matter to baryon mass ratio is (22/13) π .

22 = 2 × 11, twice the Yang-Mills integer. 13 = $|b_2'$ numerator$|$, the VL-modified SU(2) beta numerator. π from the circular geometry ($R_2 = \pi/4$).

Predicted: (22/13) π = 5.3165. Measured: 5.3204. Miss: 0.073%.

The integer ratio 22/13 is verified exact in Fraction arithmetic. The 0.073% miss is within the measurement uncertainty of the Planck 2018 values for Ω_DM and Ω_b from which the ratio is derived.

No derivation exists. The working direction: treat (22/13) π as the DM/baryon formula and test its statistical significance against random integer pools (PHYS-31).

(Backed by phys25_platform.py Section 5: 3/3 checks.)

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6. The Hubble Correction

The per-transit correction to H_0 at each soliton boundary crossing is $(1-r) = \alpha^2 \pi^2 (20/13)$.

20 = $|3b_3'|$, from the VL-modified SU(3) beta. 13 = $|6b_2'|$, from the VL-modified SU(2) beta. α^2 is the electromagnetic coupling squared (two-loop level). $\pi^2 = 32R_4$ is the 4D geometric content.

Predicted $(1-r) = 0.000809$. At N = 100 boundary transits: H_0(CMB) = 73.04 × (1 − 0.000809)^100 = 67.364 km/s/Mpc. Measured: 67.36. Miss: 0.007%.

The ratio 20/13 is exact in Fraction arithmetic. The 0.007% miss is limited by the measurement precision of H_0, not by the formula.

N = 100 is assumed, not derived. The physical mechanism connecting VP-like corrections to soliton boundary crossings is not known. The working direction: treat the per-transit formula as given, determine N from galaxy survey data (PHYS-35), and seek the physical mechanism (PHYS-34).

(Backed by phys25_platform.py Section 6: 4/4 checks.)

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7. The Baryon Density

Two candidate formulas for the baryon density parameter Ω_b.

Set A: Ω_b = $R_4 \times \alpha \times 22$ = 0.0495. Miss from measured 0.0490: 1.05%.

Set B: Ω_b = $2/(13\pi)$ = 0.04897. Miss from measured 0.0490: 0.060%.

Set B uses fewer inputs (only the integer 13 and π), achieves a tighter hit (17 × closer), and produces a cleaner downstream chain. Set B is adopted as the primary baryon formula.

From Set B, the dark matter density follows: Ω_DM = Ω_b × (DM/baryon) = $[2/(13\pi)] \times (22/13)\pi$ = 44/169. The π in the denominator of Ω_b cancels the π in the numerator of the DM ratio, leaving a pure rational. The dark matter density parameter is 44/169 = $(4 \times 11)/(13^2)$ — a ratio of the Yang-Mills integer to the square of the VL SU(2) beta numerator. No transcendentals.

(Backed by phys25_platform.py Section 7: 4/4 checks, including EXACT verification of 44/169.)

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8. The Derived Omega Chain

From two independent formulas (DM/baryon and Ω_b), the full cosmic energy budget follows.

Quantity	Formula	Predicted	Measured	Miss
Ω b	$2/(13\pi)$	0.04897	0.0490	0.06%
Ω DM	44/169	0.26036	0.2607	0.13%
Ω matter	Ω b + Ω DM	0.30933	0.3097	0.12%
Ω DE	$1 - \Omega$ matter	0.69067	0.6903	0.05%

All four density parameters within 0.15% of Planck 2018 measurements. The flat universe condition Ω_total = 1.000 is satisfied exactly by construction.

The input count: one Level 2 value (α, used only in Formulas 1, 3, and Set A Ω_b — not in Set B), four Level 1 integers (11, 13, 19, 20), and one geometric constant (π). The Set B Omega chain uses only the integer 13, the integer 11, and π . Two independent formulas produce four predictions. The system is overconstrained.

(Backed by phys25_platform.py Section 8: 4/4 checks.)

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9. The VP Step Connection

Two formulas predict similar per-transit corrections.

Formula A: $\alpha/(3\pi) = 0.000774$ (from the VP mechanism — one factor of α for the electromagnetic coupling, $1/(3\pi) = 1/(12R_2)$ for the VP step size).

Formula B: $\alpha^2 \pi^2(20/13) = 0.000809$ (from the product form scan — α^2 at two-loop level, π^2 from 4D geometry, 20/13 from modified beta ratio).

The ratio B/A = 1.044. This equals $\alpha \times 60\pi^3/13$, verified exact against mpmath.

The near-equality (4.4% difference) between two structurally different formulas suggests Formula B may be the two-loop refinement of Formula A. At one loop: the per-transit correction is $\alpha/(3\pi)$. At two loops: additional structure from the SU(3)/SU(2) beta ratio enters, giving $\alpha^2 \pi^2(20/13)$. The two-loop formula is closer to the H_0-derived target (0.08% miss vs 4.3% miss).

(Backed by phys25_platform.py Section 9: 3/3 checks.)

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10. The Combinatoric Scan

A systematic scan of ratios (p/q) × π ^b, where p and q are drawn from the beta-integer pool and b ∈ {−1, 0, 1}, found additional hits against measured quantities.

$\sin^2\theta$_W ≈ 3/13 = 0.2308. Measured: 0.2312. Miss: 0.20%. The generation count divided by the VL SU(2) beta numerator.

Ω_b ≈ 2/(13 π) = 0.04897. Measured: 0.0490. Miss: 0.06%. This became the Set B baryon formula.

Both hits use the integer 13 — the VL-modified SU(2) beta numerator. This integer appears in every cosmological formula in the series. It is the Cabibbo Doublet's signature: the integer that did not exist before the CD was added to the SM.

The scan tested approximately 12,000 combinations against 8 targets. The hit rate and quality must be compared to random integer pools of the same size and range. This comparison is the purpose of PHYS-31 — the statistical control gate for Track B.

(Backed by phys25_platform.py Section 10: 3/3 checks.)

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11. Internal Consistency

The formula set is self-consistent in three specific ways.

The π cancellation: Ω_b × (DM/baryon) = [2/(13 π)] × [(22/13) π] = 44/169. The π in the baryon formula's denominator cancels the π in the DM ratio's numerator. This is verified at 30+ digits of precision in Fraction arithmetic. The dark matter density is a pure rational.

The integer decomposition: 44 = 4 × 11 = 4 × YM. 169 = 13^2 = $|b_2$' numerator$|^2$. The dark matter density is (4 × Yang-Mills) / (VL SU(2) beta numerator squared). Both integers trace to the gauge group.

The overconstrained system: two independent formulas (DM/baryon and Ω_b) produce four predictions (Ω_b, Ω_DM, Ω_matter, Ω_DE). All four match measured values. This is not circular — the formulas were found independently by different scan methods, and the π cancellation producing a pure rational was not designed.

(Backed by phys25_platform.py Section 11: 3/3 checks, all EXACT.)

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12. The Program

Three research tracks extend from this paper.

Track A — Unification Completion. Five papers (PHYS-26 through PHYS-30) complete the Cabibbo Doublet unification program: normalization resolution, $\sin^2\theta$_W from 3/8, VL two-loop betas, GUT threshold corrections, α_s prediction. These are standard particle physics computations on the PHYS-24 ground. They proceed regardless of Track B outcomes.

Preview: $\sin^2\theta$_W = 3/8 at tree level (the SU(5) GUT prediction). Running from M_GUT = 10^15.5 GeV down to M_Z with the CD modified betas reduces 0.375 toward the measured 0.231. The full computation is PHYS-27. The preview shows the CD betas produce M_GUT in the correct range.

Track B — Beta Cosmology. Five papers (PHYS-31 through PHYS-35) investigate whether the cosmological formulas are physics or coincidence. PHYS-31 is the GATE: a statistical control comparing the beta integer pool against 10,000 random pools. If $p > 0.05$ (random pools match the beta pool's hit quality), Track B is parked. If $p < 0.01$, Track B is promoted. Papers 32–35 proceed only if the gate passes.

Track C — Structural Foundations. Two papers (PHYS-36, PHYS-37) extend the remainder framework and the Koide analysis. PHYS-36 decomposes A_3 to test whether the 87% cancellation pattern persists at three loops. PHYS-37 investigates whether QCD corrections explain the three-sector Koide amplitude ordering.

Each paper has a backing script following phys24_script_rules.md. Each has a stated abort condition. The direction is set. If it is wrong, we backtrack.

(Backed by phys25_platform.py Section 12: 2/2 checks.)

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13. What This Paper Does Not Claim

The six cosmological formulas are not derived from first principles. They were found by scanning. The claim is that the formulas exist, use only library values, and hit measured values at sub-percent precision. The claim is not that the formulas are correct physics.

The statistical significance of the hits is not established. The combinatoric scan tested thousands of combinations. Some hits are expected by chance. Whether the beta integers produce better hits than random integers is the open question addressed by PHYS-31.

The physical mechanism for the per-transit correction is not known. The formula $\alpha^2 \pi^2(20/13)$ has no derivation from vacuum polarization or any other known process. The formula is an observed numerical pattern.

The N = 100 boundary count is assumed. No galaxy survey has been consulted. The H_0 prediction at 0.007% miss uses an assumed N that may be wrong.

The interpolation fraction f = 0.44 for the cosmological constant has no formula. It is a fitted number. If it cannot be derived from the beta integers, the Λ prediction requires a free parameter.

The working direction may be wrong. Every formula may be coincidence. The abort conditions are stated for each track and each paper. If the direction fails, the PHYS-24 ground survives (it is independent of cosmology), and the series continues with Track A and Track C only.

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14. Falsification Conditions

Every operational commitment has a stated kill condition.

Commitment	What breaks it
Beta integers control Λ	Two-loop correction moves prediction away from measured
DM/baryon = (22/13) π	Future CMB measurement (CMB-S4, LiteBIRD) deviating >3σ from 5.317
H_0 from $\alpha^2 \pi^2(20/13)$	Actual boundary count N far from 100 (kills mechanism, not formula)
$\Omega_b = 2/(13\pi)$	Precision Ω_b measurement deviating >3σ from 0.04897
Statistical significance	PHYS-31 $p > 0.05$ (random pools match beta pool)
Per-transit mechanism	VP analysis cannot produce correct sign, magnitude, or ratio
Track A (unification)	$\sin^2\theta_W$ or α_s prediction >3σ from measured
Track B (cosmology)	Statistical control fails ($p > 0.05$)
Track C (structure)	A_3 shows no cancellation pattern

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15. Summary

Session 3 found that the gauge group integers control unification. Session 4 found that the same integers appear to control cosmology. The integers 13, 19, 20, and 22 — all traceable to SU(2) and SU(3) beta coefficients modified by the Cabibbo Doublet — produce seven cosmological predictions at sub-percent precision from zero cosmological input.

The predictions are staged, not proved. The formulas are found, not derived. The statistical significance is not yet established. The physical mechanism is not known. The direction is set, the abort conditions are stated, and the research program is specified.

Every number in this paper is computed and checked by phys25_platform.py. The script passes 47/47. Every integer traces to phys24_lib.py through exact Fraction arithmetic. Every comparison value traces to published measurements cited in DATA-4.

The beta coefficients run further than expected. Here is where they go.

PHYS-25: The Session 4 Operational Map. 47/47 checks. The direction is set. Published April 2, 2026. This paper is never edited after publication.

Appendix A: Complete Prediction Table

All predictions from beta integers + α + π . No cosmological input.

#	Observable	Formula	Integers Used	Predicted	Measured	Miss
1	$\log_{10}(\Lambda$ Pl) [SM]	α^(3 × 19)	3, 19	−121.80	−121.54	0.26 dec
2	$\log_{10}(\Lambda$ Pl) [VL]	(α/3 π)^(3 × 13)	3, 13, π	−121.33	−121.54	0.21 dec
3	DM/baryon	(22/13) π	11, 13, π	5.317	5.320	0.07%
4	H_0(CMB)	73.04 × (1−α^2 π^2·20/13)^100	α, 20, 13, π	67.364	67.36	0.007%
5	Ω b	2/(13 π)	13, π	0.04897	0.0490	0.06%
6	Ω DM	44/169	11, 13	0.26036	0.2607	0.13%
7	Ω matter	Ω b + Ω DM	11, 13, π	0.30933	0.3097	0.12%
8	Ω DE	1 − Ω matter	11, 13, π	0.69067	0.6903	0.05%

Appendix B: Integer Traceability

Integer	Origin	Beta Coefficient	Appears In
11	Yang-Mills: −(11/3)$C_2(G)$	b_2 gauge = −22/3, b_3 gauge = −11	DM(22), Ω b(22), Ω DM(44)
19	SM SU(2) numerator	b_2 SM = −19/6	Λ SM(57=3 × 19), identity 19/13
13	VL SU(2) numerator	b_2 mod = −13/6 = b_2 SM + 1	Λ VL(39), DM(22/13), H_0(20/13), Ω b(2/13 π), $\sin^2\theta$ W(3/13)
20	VL SU(3) numerator × 3	b_3 mod = −20/3 = −7 + 1/3	H_0(20/13)
22	2 × Yang-Mills	2 × 11	DM(22/13), Ω b($R_4\alpha$22), Ω DM(44=2 × 22)

Integer	Origin	Beta Coefficient	Appears In
3	Generations	Anomaly cancellation	Λ exponents(3×19, 3×13)

Appendix C: The Research Program

Paper	Track	Title	Gate	Abort Condition
PHYS-26	A	Normalization Resolution	None	Documentation task
PHYS-27	A	$\sin^2\theta$ W from 3/8	PHYS-26	>5% miss from 0.23122
PHYS-28	A	VL Two-Loop Betas	PHYS-26	VL correction worsens Δ
PHYS-29	A	GUT Thresholds	PHYS-28	M T/M X > 100 (fine-tuned)
PHYS-30	A	α s Prediction	PHYS-28,29	>3σ from 0.1180
PHYS-31	B	Statistical Control	None	**p > 0.05 kills Track B**
PHYS-32	B	Set B Omegas	PHYS-31	Neither set uniformly better
PHYS-33	B	Λ Interpolation	PHYS-31	No formula for f, two-loop wrong direction
PHYS-34	B	Per-Transit Mechanism	PHYS-31	VP cannot produce formula
PHYS-35	B	Boundary Count	PHYS-31,34	N far from 100
PHYS-36	C	A_3 Decomposition	None	No cancellation pattern
PHYS-37	C	Koide Amplitude	None	QCD correction wrong ordering

Appendix D: Verification Summary

Script	Checks	Status
phys25 platform.py	47/47	PASS
beta unification test.py	15/15	PASS
qed predicts gr.py	10/10	PASS
qed gr scan 2.py	10/10	PASS
phys24 lib.py self-test	21/21	PASS
phys24 lib test.py	148/148	PASS
8 PHYS-24 demo scripts	62/62	PASS
6 Session 3 scripts	98/98	PASS
Grand total	**411/411**	**ZERO FAILURES**

Appendix E: The Six Formulas — Exact Specification

Formula 1a: $\log_{10}(\Lambda$_Planck$) = 57 \times \log_{10}(\alpha)$, where $57 = 3 \times |\text{numerator}(6 \times b_2\text{\_SM})| = 3 \times 19$.

Formula 1b: $\log_{10}(\Lambda$_Planck$) = 39 \times \log_{10}(\alpha/(3\pi))$, where $39 = 3 \times |\text{numerator}(6 \times b_2\text{\_mod})| = 3 \times 13$.

Formula 2: DM/baryon = $(2 \times 11 / 13) \times \pi = (22/13)\pi$, where 11 = Yang-Mills, 13 = $|\text{numerator}(6 \times b_2\text{\_mod})|$.

Formula 3: $(1-r) = \alpha^2 \times \pi^2 \times (20/13)$, where $20 = |3 \times b_3\text{\_mod}|$, $13 = |6 \times b_2\text{\_mod}|$. $H_0(\text{CMB}) = H_0(\text{local}) \times (1-r)$^N.

Formula 4 (Set B): Ω_b $= 2/(13\pi)$, where 13 = $|\text{numerator}(6 \times b_2\text{\_mod})|$.

Formula 5 (derived): Ω_DM = Ω_b $\times$ DM/baryon = $[2/(13\pi)] \times [(22/13)\pi] = 44/169 = (4 \times 11)/(13^2)$.

Identity: $57/39 = 19/13 = |\text{numerator}(6 \times b_2\text{\_SM})| / |\text{numerator}(6 \times b_2\text{\_mod})|$. Exact in Fraction arithmetic.

Supporting Appendix Tables for PHYS-25

TABLE 25.1: THE COMPLETE PREDICTION TABLE — SET B PRIMARY

#	Observable	Formula	Predicted	Measured	Abs Miss	Rel Miss	Script Check
1a	$\log_{10}(\Lambda$ Pl) [SM]	$57 \times \log_{10}(\alpha)$	−121.800	−121.54	0.260 dec	0.21%	S4.1 PASS
1b	$\log_{10}(\Lambda$ Pl) [VL]	$39 \times \log_{10}(\alpha/(3\pi))$	−121.333	−121.54	0.207 dec	0.17%	S4.2 PASS
1c	$\log_{10}(\Lambda$ Pl) [mean]	average of 1a,1b	−121.566	−121.54	0.026 dec	0.02%	derived
2	DM/baryon	$(22/13)\pi$	5.3165	5.3204	0.0039	0.073%	S5.1 PASS
3	(1−r) per transit	$\alpha^2 \pi^2 (20/13)$	0.000809	0.000809	6.6×10^{-7}	0.082%	S6.1 PASS
4	H_0(CMB)	$73.04 \times r^{100}$	67.364	67.36	0.004	0.007%	S6.2 PASS
5	Ω b	$2/(13\pi)$	0.04897	0.0490	0.00003	0.060%	S7.2 PASS
6	Ω DM	44/169	0.26036	0.2607	0.00034	0.132%	S8.1 PASS
7	Ω matter	Ω b + Ω DM	0.30933	0.3097	0.00037	0.121%	S8.2 PASS
8	Ω DE	1 − Ω matter	0.69067	0.6903	0.00037	0.054%	S8.3 PASS
9	$\sin^2\theta$ W	3/13	0.23077	0.23122	0.00045	0.195%	S10.1 PASS

Nine predictions. Maximum miss: 0.26 decades (Λ , over 122 orders of magnitude). All others within 0.2%.

TABLE 25.2: THE INPUT SET — EVERYTHING THAT ENTERS THE FORMULAS

Input	Value	Type	Source	Used In Formulas
α	1/137.035999177	Level 2 (measured)	DATA-4 B1	1a, 1b, 3, 4, Set A Ω b
11	Yang-Mills integer	Level 1 (gauge)	$-(11/3)C_2(G)$	2(22), 5(Set A: 22), 6(44)

Input	Value	Type	Source	Used In Formulas
13	$\lvert b_2$ mod numerator$\rvert$	Level 1 (gauge + CD)	b_2 mod = −13/6	1b(39), 2(22/13), 3(20/13), 5(2/13 π), 6(169), 9(3/13)
19	$\lvert b_2$ SM numerator$\rvert$	Level 1 (gauge)	b_2 SM = −19/6	1a(57), identity 19/13
20	$\lvert 3 \times b_3$ mod$\rvert$	Level 1 (gauge + CD)	b_3 mod = −20/3	3(20/13)
π	3.14159…	Level 1 (geometric)	$R_2 = \pi/4$	1b, 2, 3, 4, 5, 7, 8
3	N gen	Level 1 (anomaly cancel.)	SM generations	1a(57=3 × 19), 1b(39=3 × 13), 9(3/13)
R_4	$\pi^2/32$ = 0.30843	Level 1 (geometric)	MATH-5	Set A Ω b only
N = 100	Assumed boundary count	**Assumed**	Not measured	4 (H_0 prediction)
H_0 local = 73.04	Local Hubble	Level 2 (measured)	SH0ES 2022	4 (H_0 prediction)

Count: 1 measured coupling (α) + 5 Level 1 integers (11, 13, 19, 20, 3) + 1 geometric constant (π) + 1 assumed parameter (N) + 1 measured reference (H_0_local). The Set B Omega chain uses only {13, 11, π } — three inputs for four outputs.

TABLE 25.3: FORMULA-BY-FORMULA INTEGER TRACEABILITY

Formula	Expression	Integer 1	Origin 1	Integer 2	Origin 2	Other
Λ (SM)	α^57	57 = 3 × 19	N gen × $\lvert b_2$ SM num$\rvert$	—	—	α
Λ (VL)	(α/3 π)^39	39 = 3 × 13	N gen × $\lvert b_2$ mod num$\rvert$	—	—	α, π
DM/baryon	(22/13) π	22 = 2 × 11	2 × YM	13	$\lvert b_2$ mod num$\rvert$	π
(1−r)	$\alpha^2 \pi^2$(20/13)	20	$\lvert 3 \times b_3$ mod$\rvert$	13	$\lvert b_2$ mod num$\rvert$	α, π

Formula	Expression	Integer 1	Origin 1	Integer 2	Origin 2	Other
Ω b (Set B)	2/(13 π)	13	|b_2 mod num|	—	—	π
Ω DM	44/169	44 = 4 × 11	4 × YM	169 = 13^2	|b_2 mod num|^2	—
$\sin^2\theta$ W	3/13	3	N gen	13	|b_2 mod num|	—
Identity	57/39	19	|b_2 SM num|	13	|b_2 mod num|	—

The integer 13 appears in 7 of 8 formulas. It is the dominant integer of the cosmological extension. It exists ONLY because the Cabibbo Doublet was added (b_2_SM + Δb_2 = −19/6 + 1 = −13/6).

TABLE 25.4: THE 19→13 TRANSFORMATION — COMPLETE IMPACT

Domain	SM value (uses 19)	CD value (uses 13)	Change	Consequence
b_2 numerator (× 6)	19	13	−6	SU(2) running weakens
b_3 numerator (× 3)	21	20	−1	SU(3) running weakens slightly
b_1 numerator (× 30)	123	125	+2	U(1) running strengthens slightly
Gap ratio	218/115 = 1.896	38/27 = 1.407	−0.489	Unification enabled
Gap distance from measured	0.538 (40%)	0.049 (3.6%)	−0.489	11 × improvement
M GUT	10^13.8 GeV	10^15.5 GeV	+1.7 dec	Proton decay testable
τ proton	~10^30 yr (excluded)	~10^34–35 yr	+4–5 dec	Hyper-K window
Λ exponent	57 (= 3 × 19)	39 (= 3 × 13)	−18	VL Λ prediction
DM denominator	—	13	new	DM/baryon = (22/13) π
H_0 denominator	—	13	new	(1−r) = α^2 π^2(20/13)
Ω b denominator	—	13 π	new	Ω b = 2/(13 π)

Domain	SM value (uses 19)	CD value (uses 13)	Change	Consequence
Ω DM denominator	—	$169 = 13^2$	new	Ω DM = 44/169
$\sin^2\theta$ W denominator	—	13	new	$\sin^2\theta$ W $\approx$ 3/13

The last five rows are cosmological consequences that emerge ONLY after the CD is added. They do not exist in the SM.

TABLE 25.5: SET A vs SET B — DETAILED COMPARISON

Quantity	Set A Formula	Set A Pred	Set A Miss	Set B Formula	Set B Pred	Set B Miss	Ratio A/B
Ω b	$R_4 \times \alpha \times 22$	0.04952	1.051%	$2/(13\pi)$	0.04897	0.060%	17.5 ×
Ω DM	$R_4 \times \alpha \times 22 \times (22/13)\pi$	0.26325	0.978%	44/169	0.26036	0.132%	7.4 ×
Ω matter	sum	0.31276	0.989%	sum	0.30933	0.121%	8.2 ×
Ω DE	1 − sum	0.68724	0.444%	1 − sum	0.69067	0.054%	8.2 ×
Property	Set A	Set B	Winner				
—	—	—	—				
Inputs for Ω b	R_4, α, 22 (3 inputs)	13, π (2 inputs)	**B** (simpler)				
α dependence	Yes (in Ω b)	No (in Ω b)	**B** (fewer L2 inputs)				
Ω DM transcendental content	Irrational (involves R_4, α, π)	Rational: 44/169	**B** (cleaner)				
Best single miss	0.444% (Ω DE)	0.054% (Ω DE)	**B** (8 × better)				

Quantity	Set A Formula	Set A Pred	Set A Miss	Set B Formula	Set B Pred	Set B Miss	Ratio A/B
Worst single miss	1.051% (Ω b)	0.132% (Ω DM)	**B** (8 × better)				
π cancellation	Does not cancel	Cancels exactly in Ω DM	**B** (structural)				

Set B is uniformly superior on every metric.

TABLE 25.6: THE EXACT ALGEBRAIC IDENTITIES

Identity	LHS	RHS	Verification	Script Check
57/39 = 19/13	Fraction(57,39)	Fraction(19,13)	EXACT	S2.7
$20/13 = \lvert b_3$ mod$\rvert/\lvert b_2$ mod num$\rvert$	Fraction(20,13)	b3 mod num/b2 mod num	EXACT	S2.8
$22/13 = (2 \times$ YM$)/\lvert b_2$ mod num$\rvert$	Fraction(22,13)	2 × YM/b2 mod num	EXACT	S5.2
$44/169 = (4 \times$ YM$)/\lvert b_2$ mod num$\rvert^2$	Fraction(44,169)	4 × YM/(b2 mod num^2)	EXACT	S11.2, S11.3
Ω b × DM/b = 44/169	$[2/(13\,\pi)] \times [(22/13)\,\pi]$	44/169	EXACT (π cancels)	S11.1
$\Delta b_2/\Delta b_1 = 15$	db2 VL/db1 VL	Fraction(15)	EXACT	phys24 CD script
Gap SM = 218/115	(b1 SM−b2 SM)/(b2 SM−b3 SM)	Fraction(218,115)	EXACT	S1.1
Gap VL = 38/27	(b1 mod−b2 mod)/(b2 mod−b3 mod)	Fraction(38,27)	EXACT	S1.2

Eight exact identities. All verified in Python Fraction arithmetic. None is numerical — all follow from the library values through exact cancellation.

TABLE 25.7: THE RESEARCH PROGRAM — DEPENDENCY AND ABORT MAP

Paper	Track	Depends On	Gate?	Script (est. lines)	Abort Condition	What Survives Abort
PHYS-26	A	—	No	phys26 normalization.py (~50)	None (documentation)	Everything
PHYS-27	A	PHYS-26	No	phys27 sin2tw.py (~40)	$\sin^2\theta$ W >5% miss	Gap ratio, CD identification
PHYS-28	A	PHYS-26	No	phys28 vl twoloop.py (~100)	VL correction worsens Δ	One-loop result stands
PHYS-29	A	PHYS-28	No	phys29 gut thresholds.py (~80)	M T/M X > 100	CD survives, min SU(5) weakened
PHYS-30	A	PHYS-28,29	No	phys30 alpha s.py (~60)	α s >3σ from 0.1180	CD survives, unification interp. weakened
PHYS-31	**B**	—	**YES**	phys31 stat control.py (~120)	**p > 0.05**	**Track B parked. 57/39=19/13 survives.**
PHYS-32	B	PHYS-31	No	phys32 set b omega.py (~80)	Neither set uniformly better	Individual formulas may survive
PHYS-33	B	PHYS-31	No	phys33 lambda interp.py (~80)	No f formula + wrong 2-loop direction	Λ prediction weakened

Paper	Track	Depends On	Gate?	Script (est. lines)	Abort Condition	What Survives Abort
PHYS-34	B	PHYS-31	No	phys34 per transit.py (~100)	VP cannot produce formula	Formula survives as pattern, not physics
PHYS-35	B	PHYS-31,34	No	phys35 boundary count.py (~100)	N far from 100	DM and Ω formulas survive
PHYS-36	C	—	No	phys36 a3 decomp.py (~80)	No cancellation pattern	A_2 anatomy stands alone
PHYS-37	C	—	No	phys37 koide amp.py (~80)	QCD wrong ordering	Koide amplitude remains open

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TABLE 25.8: THE INTEGER 13 — COMPLETE APPEARANCE LIST

Formula/Identity	How 13 Appears	Algebraic Form	Numerical Role
b_2 mod	−13/6	b_2 SM + Δb_2 = −19/6 + 1	Modified SU(2) running
Gap ratio (VL)	38/27	(b_1 mod−b_2 mod)/(b_2 mod−b_3 mod)	Contains 13 in b_2 mod
Λ exponent (VL)	39 = 3 × 13	N gen × |b_2 mod num|	Controls Λ scale
DM/baryon	22/13	(2 × YM)/|b_2 mod num|	Denominator of ratio
Per-transit H_0	20/13	|3b_3 mod|/|b_2 mod num|	Denominator of ratio
Ω b (Set B)	2/(13 π)	2/(|b_2 mod num| × π)	Denominator
Ω DM	$44/169 = 44/13^2$	(4 × YM)/|b_2 mod num|^2	Denominator squared

Formula/Identity	How 13 Appears	Algebraic Form	Numerical Role
$\sin^2\theta$ W scan	3/13	N gen/$\lvert b_2$ mod num$\rvert$	Denominator
Exact identity	19/13	$\lvert b_2$ SM num$\rvert$/$\lvert b_2$ mod num$\rvert$	Ratio of SM to VL
VP ratio B/A	$\alpha \times 60\,\pi^3/13$	see Section 9	Denominator

10 appearances. The integer 13 is the Cabibbo Doublet's fingerprint on cosmology. It enters through b_2_mod = −13/6, which exists only because $\Delta b_2 = 1$ from the (3,2,1/6) representation.

TABLE 25.9: WHAT THE CABIBBO DOUBLET PROVIDES TO EACH DOMAIN

Domain	What CD Provides	Level	Tested By
Gauge unification	Gap ratio 38/27, distance 0.049	1	sin2 theta w 1.py 9/9
Proton decay	M GUT = 10^15.5, $\tau \sim$ 10^34–35	Derived	phys24 cabibbo doublet.py 10/10
CKM physics	Extended 3×4 matrix, $\lvert V$ ub'$\rvert \approx$ 0.045	2	Web-verified citations
LEP anomaly	Z-b-b vertex correction via θ_{34}	2	Web-verified citations
Higgs physics	Loop contribution to gg→H	2	Web-verified citations
Cosmological constant	Integer 13 in Λ exponent ($39 = 3 \times 13$)	Proposed	S4.1–S4.4 PASS
Dark matter fraction	Integer 13 in DM ratio (22/13)	Proposed	S5.1–S5.3 PASS
Hubble constant	Integers 20,13 in per-transit (20/13)	Proposed	S6.1–S6.4 PASS
Baryon density	Integer 13 in Ω b = $2/(13\,\pi)$	Proposed	S7.1–S7.4 PASS
Dark energy	Via Ω chain from above	Proposed	S8.1–S8.4 PASS
Weak mixing angle	Integer 13 in $\sin^2\theta$ W $\approx$ 3/13	Proposed	S10.1 PASS

The first five rows are from PHYS-15 through PHYS-20 (established). The last six rows are from Session 4 (proposed).

TABLE 25.10: THE VP STEP — TWO FORMULAS COMPARED

Property	Formula A	Formula B
Expression	$\alpha/(3\pi)$	$\alpha^2 \pi^2(20/13)$
Value	0.000774	0.000809
Physical picture	One VP step: α coupling $\times$ $1/(3\pi)$ step size	Two-loop: $\alpha^2 \times$ 4D geometry $\times$ beta ratio
Loop order	One-loop	Two-loop
Beta content	None (pure QED)	20 (SU(3)) and 13 (SU(2))
Miss from H_0 target	4.32%	0.082%
Ratio B/A	1.044	—
Ratio formula	$\alpha \times 60\pi^3/13$	—
Interpretation	VP mechanism prototype	VP mechanism + gauge structure correction

Formula B includes gauge structure (the 20/13 beta ratio) that Formula A lacks. The 60 π^3 in the ratio is 60 $\times$ 31.006 = 1860.4; divided by 13 = 143.1; times α = 143.1/137.04 = 1.044. The near-unity of the ratio may reflect that the gauge correction to the VP step is of order α, as expected for a one-loop correction to a one-loop process.

TABLE 25.11: THE π CANCELLATION IN DETAIL

Step	Expression	Contains π ?	Fraction form
Ω b formula	$2/(13\pi)$	Yes (denominator)	2/(13 $\times$ pi f)
DM/baryon formula	$(22/13)\pi$	Yes (numerator)	(22 $\times$ pi f)/13
Product: Ω DM	$[2/(13\pi)] \times [(22/13)\pi]$	**No —** π cancels	(2 $\times$ 22)/(13 $\times$ 13) = 44/169
Simplified	44/169	Pure rational	Fraction(44, 169)
Decomposition	$(4 \times 11)/(13^2)$	4 $\times$ YM / (b_2 mod num)^2	Exact

The cancellation is verified in phys25_platform.py at 30+ digit precision (check S11.1: EXACT). The dark matter density parameter is a ratio of small integers from the gauge group. No transcendentals.

TABLE 25.12: EXPERIMENTAL FALSIFICATION TIMELINE

Falsification Target	Experiment	When	What Would Falsify	What Survives
CD direct production	HL-LHC	Now–2040	Exclusion above 6 TeV	Gap ratio survives (L1 arithmetic)
CKM deficit	Belle II	Now–2030+	Deficit disappears	Gap ratio survives
Proton decay	Hyper-K	2027–2037	τ > 10^35 (min SU(5) excluded)	CD + anomalies survive
DM/baryon = (22/13) π	CMB-S4, LiteBIRD	~2030+	Ratio deviates >3σ from 5.317	Other formulas independent
Ω b = 2/(13 π)	Precision BBN + CMB	Ongoing	Ω b deviates >3σ from 0.04897	DM and Λ formulas independent
H_0 convergence	SH0ES + Planck successor	Ongoing	H_0 values diverge further	DM and Ω formulas independent
Statistical significance	PHYS-31	Next session	$p > 0.05$	Track A, exact identities
Λ from β exponents	PHYS-33 two-loop test	Next session	Two-loop moves wrong direction	Other formulas independent

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TABLE 25.13: THE COMPLETE VERIFICATION LEDGER

Script	Session	Checks	Status	What It Backs
phys25 platform.py	4	47/47	PASS	**This paper (PHYS-25)**
beta unification test.py	4	15/15	PASS	Beta cosmology discovery
qed predicts gr.py	4	10/10	PASS	QED-to-GR scan 1
qed gr scan 2.py	4	10/10	PASS	QED-to-GR scan 2
phys24 lib.py self-test	4	21/21	PASS	Platform library

Script	Session	Checks	Status	What It Backs
phys24 lib test.py	4	148/148	PASS	Full DATA-4 verification
phys24 gap ratio.py	4	5/5	PASS	Gap ratio anatomy
phys24 democracy.py	4	10/10	PASS	Boson problem
phys24 cabibbo doublet.py	4	10/10	PASS	CD specification
phys24 two loop.py	4	8/8	PASS	Two-loop improvement
phys24 koide status.py	4	10/10	PASS	Koide C_3 closure
phys24 a2 anatomy.py	4	7/7	PASS	A_2 decomposition
phys24 pslq null.py	4	4/4	PASS	PSLQ null + sanity
phys24 data4 check.py	4	8/8	PASS	DATA-4 consistency
sin2 theta w 1.py	3	9/9	PASS	Gap ratios, enumeration
a 2 decomposition 0.py	3	7/7	PASS	A_2 three-piece
bessel pslq 0.py	3	6/6	PASS	Bessel independence
data 2 to 3 test 1.py	3	32/32	PASS	DATA-3 consistency
data 4.py	3	38/38	PASS	DATA-4 consistency
unification test.py	3	6/6	PASS	Two-loop ODE integration
GRAND TOTAL	**ALL**	**411/411**	**ZERO FAILURES**	**Complete series**

TABLE 25.14: THE THREE POSSIBLE OUTCOMES

Outcome	Track A Result	Track B Result	Interpretation	Series Status
A: Full Success	$\sin^2\theta$ W and α s match	$p < 0.01$, mechanism found	β integers control unification AND cosmology	Unified framework, 40 orders of magnitude
B: Partial Success	$\sin^2\theta$ W and α s match	$p > 0.05$ OR no mechanism	CD fixes unification; cosmology is coincidence	Particle physics story only
C: Minimal Success	$\sin^2\theta$ W or α s wrong	$p > 0.05$	Gap ratio arithmetic correct; physical interpretation weakened	Mathematical anatomy + methodology

Each outcome is informative. None is wasted. The series learns from every result including the nulls. The methodology (exact Fraction arithmetic, verified scripts, Level 1/Level 2 boundary, falsification conditions) survives all three outcomes.

TABLE 25.15: THE FORMULA SET — WHAT USES WHAT (MATRIX VIEW)

	α	π	11	13	19	20	3	R_4	N	H_0 loc
** Λ (SM)**	●				●		●			
** Λ (VL)**	●	●		●			●			
DM/baryon		●	●	●						
(1−r)	●	●		●		●				
H_0(CMB)	●	●		●		●			●	●
Ω b (Set B)		●		●						
Ω DM			●	●						
Ω matter		●	●	●						
Ω DE		●	●	●						
$\sin^2\theta$ W				●			●			
57/39 identity				●	●					
Count	4	7	4	**10**	2	2	3	0	1	1

The integer 13 appears in 10 of 11 formulas. π appears in 7. α appears in 4. R_4 appears in 0 (Set B eliminates it from the Ω chain). The Cabibbo Doublet's modification of b_2 from −19/6 to −13/6 is the single change that enables the entire cosmological extension.

End of supporting appendix tables for PHYS-25. 15 tables. Every number traces to phys25_platform.py (47/47 PASS) or to prior verified scripts. Every integer traces to the gauge group through the beta coefficients. The tables document the discovery, the formulas, the program, the abort conditions, and the verification chain. Grand total: 411/411 checks across 20 scripts in 2 sessions. Zero failures.

[**HOWL-PHYS-25-2026**] The Session 4 Operational Map — From Gauge Integers to Cosmological Predictions. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-25-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-21-2026**] The Level 1 / Level 2 Boundary — From Observed Structure to Unobserved Particles. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-21-2026

[**HOWL-PHYS-24-2026**] The Session 3 Operational Lexicon. Github: https://github.com/ghowland/papers/tree/main/pape
PHYS-24-2026